

Test Math Notes

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Chapter 1

Introduction

1.1 Example

Definition 1.1.1. A random variable X is Gaussian if $X \sim \mathcal{N}(\mu, \sigma^2)$.

Theorem 1.1.1. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$. Then $\bar{X}$ is unbiased.

Proof. By linearity, $\mathbb{E}[\bar{X}] = \mu$. □

1.2 Algorithm Example

Algorithm 1 Gradient Descent

```
1: Initialize  $x_0$ 
2: for  $k = 0, 1, 2, \dots$  do
3:    $x_{k+1} \leftarrow x_k - \eta \nabla f(x_k)$ 
4: end for
```

1.3 Code Example

```
1 import numpy as np
2 import pandas as pd
3 import scipy as
4 def gradient_descent(x0, eta, grad, steps):
5     x = x0
6     for _ in range(steps):
7         x = x - eta * grad(x)
8     return x
```